Rhea Madhogarhia
Navneet Bhasin
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Bio-Weapons and Bio-Terrorism: Then to Now

Infecting over 700 million people and killing over 7 million people worldwide, the COVID-19 pandemic was unsurprisingly labeled a "Public Health Emergency of International Concern" by the World Health Organization (WHO). COVID-19 is caused by severe acute respiratory syndrome coronavirus 2 (SARS-CoV-2). The global spread of COVID-19 can be credited to the virus's ease of transmission from person to person and its harmful and fatal nature can be attributed to "the widespread expression of SARS-CoV-2 host cell entry factors across human tissue types", the dysregulation of cardiovascular function, and a hyperactive innate immune response to infection (Kumar et al.). Given the extremely harmful impact COVID-19 had on the global population certain terrorist and extremist groups advocated for the intentional spread of COVID-19: "right-wing extremist groups, like CoronaWaffen, explicitly asked their followers to spread the virus by coughing on their local minority or by attending to specific places where religious or racial minorities gather" (UNICRI). While the virus was not introduced to the human population intentionally, CoronaWaffen's attempts to deliberately spread the virus to harm or kill a part of the population were attempts to adapt COVID-19 into a biological weapon and commit acts of terrorism (UNICRI). According to WHO, bioweapons are "microorganisms like virus, bacteria or fungi, or toxic substances produced by living organisms that are produced and released deliberately to cause disease and death in humans, animals or plants" (WHO). There are roughly five types of bioweapons: bacteria, rickettsiae, viruses, fungi, and toxins. Given that COVID-19 is a virus, if the coronavirus had been intentionally introduced into the human population, it easily would have been considered a biological weapon.

Remembering the millions of deaths and illnesses that COVID-19 caused, it is not hard to imagine just how destructive biological weapons can be: some viruses can spread easily through airborne droplets, fungi can be widely disseminated as herbicidal warfare over crop fields, and bacteria and rickettsia can infect people through a variety of delivery systems (Schneider). Bioweapons have historically been used for military applications, assassinations, targeted illness, and causing food shortages, and have the potential to inspire economic loss, global illness, and environmental catastrophe; its categorization as a weapon of mass destruction seems fitting (Schneider). While we've already discussed extremist groups seizing an existing pandemic as a bioweapon, in this paper we will discuss a bacterial disease that has been historically used as a bioweapon: anthrax.

Anthrax is a zoonotic disease caused by the bacterium *Bacillus anthracis (*BA*)*. While BA is found naturally in soil and is a common disease found in livestock and wild animals, humans can contract anthrax through ingestion of meat from an infected animal, the inhalation of anthrax spores, injection directly into the body, or exposure to spores through cuts or scrapes on the skin (CDC). The CDC has labeled BA as a Tier 1 biological agent (agents that present the greatest risk of deliberate misuse and potential to devastate infrastructure, economy, and public safety) because of anthrax's severity and fatality as a disease, its ability to be discreetly packaged and delivered, and its success as a bioweapon historically, which we will discuss more later (CDC).

Anthrax can affect the body in three ways: cutaneous, gastrointestinal, or inhalational infection. When BA spores are inhaled, the spores collect in the alveoli of the lungs where an immune response kicks in causing macrophages to attack and consume the spores. The spore coat around the bacteria keeps it protected from being destroyed by the macrophages and the macrophages end up transporting BA to the host body's lymph nodes. The lymphatic system provides a nutrient-rich environment for the spores to spread, grow, and enter the bloodstream (NIH). Once in the bloodstream, anthrax can spread to multiple organs throughout the body and produce toxins and proteins. BA's "lethal factor" protein results in the loss of cellular regulation and ion metabolism and BA's "edema factor" protein causes the cleaving of protein transduction pathways, leading to cell death or damaged immune responses. These effects make inhalational anthrax extremely deadly and, even if antibiotics are successful in killing the bacterial infection, the effects and damage created by BA persist: an infected patient may die even after being treated by antibiotics. Gastrointestinal (GI) anthrax occurs from ingesting contaminated meat, bringing spores to the GI tract where bacterial replication causes bleeding and lesions called mucosal ulcerations. Finally, cutaneous anthrax occurs when a break or cut in the skin allows anthrax spores to directly enter the bloodstream where they may circulate to the lungs, kidney, or spleen (NIH).

So, what sets apart BA from other bacteria in the *Bacillus* genus? Researchers have identified a mutation in BA's global regulator *plcR* gene, a gene that normally controls the transcription of virulence factors in other *Bacillus* species. When this gene is dysfunctional, BA likely relies more heavily on its plasmid-encoded virulence factors (the edema and lethal factor toxins); when the expression of these factors is uncontrolled there is the danger of overexpression of BA's toxins contributing to the severity of illness that BA can cause (NIH).

It's clear now that BA's residence in a human body is fatal to its host. While anthrax is an occupational risk in professions that deal with raw meat, animal hide or fur, or infected animals, inhalational anthrax is rare. Cases of inhalational anthrax sound the alarm for bioterrorism, especially considering the use of anthrax in bioterrorism attacks through history. Although, generally, the use of biological agents as terrorist weapons is rare, anthrax is the most commonly used agent in a bioterrorism event. A majority of reported anthrax attacks occurred in the US by a single perpetrator in 2001 (Tin, Sabeti, Ciottone). The 2001 anthrax attacks were characterized by anthrax-laced letters being sent to news outlets and congressional office buildings in which 5 of the 22 infected people died. Occurring right after the devastating events of 9/11 that same year, the letters garnered serious attention as evidence of another terrorist attack on the US. Billions of dollars were spent on decontaminating post offices and government buildings and parts of the medical community put serious effort into studying the effects of anthrax and how to protect people from it. While anthrax was not a new disease in 2001, few doctors were experienced with treating anthrax in humans, and conducting studies on anthrax in humans is not feasible or ethical (Bock). So, to roll out antibiotics to treat a zoonotic disease such as anthrax, the FDA declared the Animal Rule which allows for the approval of drugs and biological products when human clinical efficacy trials are not possible (FDA). Dr. Mary Wright of the National Institute of Allergy and Infectious Diseases notes that the anthrax letter attacks, "revealed the urgent need to be able to understand old pathogens using current modern medical tools", which research infrastructure updates, like the Animal Rule, aimed to address (Bock).

While the 2001 anthrax attacks targeted individuals through exposure to BA, the fatality of exposure to BA makes anthrax a good candidate for violent actors and a serious military and bioterrorism threat (NIH). Letters addressed to individuals is a somewhat contained delivery system, but other bio-weapon delivery systems, such as aerial spraying or explosives, can vastly increase the spread of a bioweapon agent. Aerosolized BA as a bioweapon could incur millions of fatalities if released over an urban area (NIH). In fact, an accidental release of aerosolized BA in 1979 at a former Soviet military facility resulted in many human and animal deaths (PBS).

This outbreak piqued international interest since the 1972 Biological Weapons Convention banned the development of biological weapons given the use of biological and chemical weapons in World War II (UNODA). The catastrophic use and potential of bioweapons for deliberately killing subsets of the population can be traced back to the 12th century but military bioweapon programs were most prevalent in the 20th century (Dowdeswell). For example, Japan conducted multiple plague dissemination attacks on Chinese cities in the 1930s and 1940s, the US developed herbicidal bioweapons to destroy rice and wheat fields during the Cold War and WWII, and the US Navy tested San Francisco's susceptibility to a mistakenly harmless bioweapon decoy during Operation Sea Spray in 1950 (Thompson; Guilleman). While military applications shouldn't be developed today, bioterrorism is still a salient threat that is only being exacerbated by recent developments in biotechnology.

Developments in gene editing technology, like CRISPR-Cas9, artificial intelligence, and machine learning have the potential to make bioterrorist attacks even deadlier and easier to conduct. Gene editing technology opens up the possibility for the creation of new biological weapons and the enhancements of old ones. The aforementioned Soviet military release of aerosolized BA was a non-CRISPR genetically engineered version of the bacteria. Today, the changes the Soviet military made to the bacteria are much easier to make given technological advancement (Aken, Hammond). CRISPR technology also makes it possible to increase a pathogen's deadliness and virality (Charlet). Furthermore, CRISPR is more cost-effective than previous gene editing techniques and a technique available for public use (Cropper). AI chatbots can give detailed instructions on how to construct novel toxins or bioweapons and gene editing kits are available online (Donaldson).

Discussions surrounding bioweapons, like the ones that took place at the Biological Weapons Convention, never fail to raise concern about their creation and use. The risk of using bioweapons to kill certain populations or to discriminate based on genetics increases as this technology becomes more easily accessible. These potential acts of bioterrorism could have global catastrophic consequences and the inciting of a pandemic could impact the world even more than a naturally occurring pandemic like COVID-19. While these risks are hypothetical and hard to imagine given the unknown makeup of novel toxins or pathogens that could be developed, we should continue work, on a public health infrastructure level, to mitigate the spread of infectious diseases. While we can try and prevent governments from using these powerful weapons, bioterrorism is unpredictable, and serious conversations about what kind of biotechnological advancements should be made accessible to the public should be had. During the anthrax attacks, we were largely uninformed about the disease and treatment, which led to measures like the Animal rule and research on the disease. Especially as countries like the United States begin to see more accounts of political violence in the news, the risk of bioterrorism is one we should seriously consider and medically inform ourselves about.

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